MIL-P-14785B(PA)
23 Nov. 1973
SUPERSEDING
MIL-P-14785A(MU)
13 January 1972

MILITARY SPECIFICATION

PROJECTILE, 60MM
METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Projectile, Metal Parts Assembly for Cartridge, 60MM, HE, M49A3 and M49A4.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-Ammunition and Special Weapons; General Specification for
MIL-M-11472	-Magnetic Particle Inspection; Process for Ferromagnetic Materials

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-410	-Qualification of Inspection Personnel (Magnetic Particle and Penetrant)
MIL-STD-414	-Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-453	-Inspection, Radiographic
MIL-STD-1168	-Lot Numbering of Ammunition
MIL-STD-1235	-Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

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DRAWINGS

ARMY

7548319 -Cartons, Ammunition for Interplant Shipment of Projectiles, Empty
9241480 -Body (Marking)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTM E8 -Tension Testing of Metallic Materials

ASTM E10 -Brinell Hardness of Metallic Materials

ASTM E192-Standard Reference Radiographs of Investment Steel Castings

Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103.

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Projectile.-The projectile shall comply with all requirements specified on drawing (dwg.) 9241480, and with all requirements specified in applicable specifications.

3.3 Soundness.-The soundness of castings shall meet the requirements of 3.3.1 and 3.3.2.

3.3.1 Surface indications.-The body casting, after machining, shall be free of those defects which would be detectable as normally viewed without the aid of non-destructive test techniques and magnetic particle indications greater than 1.8 inch in length. Surface indications shall also comply with the requirements of 3.3.1.1 and 3.3.1.2. Testing shall be in accordance with 4.4.2.1.

3.3.1.1 Interior surface indications

3.3.1.1.1 Indications shall not exist in the tail section, within the ".58 Ref. Mean Tangent Spherical R.", area.

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3.3.1.1.2 Indications shall not extend over more than two threads and must be less than the full depth of the threads.

3.3.1.1.3 No sharp protrusions are acceptable.

3.3.1.2 Exterior surface indications.

3.3.1.2.1 Indications on the face of the body, at the fuze end, shall not extend through the body wall or extend more than 1/16 inch into the inside or outside diameter.

3.3.1.2.2 The bourrelet shall be free of any area of "as cast" finish.

3.3.2 Radiographic indications.-There shall be no evidence of cracks, cold shuts or hot tears. All other radiographic indications shall be categorized as either smooth edged (round or elongated) or irregularly shaped. Smooth edged and irregularly shaped indications shall comply with the requirements of 3.3.2.1, 3.3.2.2, 3.3.2.3, and 3.3.2.4. Testing shall be in accordance with 4.4.2.2.

3.3.2.1 Section A (see dwg. 9207928, covering a detail of dwg. 9241480).

3.3.2.1.1 No individual smooth edged indication, in Section A, shall exceed the area of the largest hole depicted in the gas hole plate for 1/8 inch thick section view #6 of ASTM E192.

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3.3.2.1.2 No individual irregularly shaped indication, in Section A, shall exceed the area of the largest indication depicted in the foreign material less dense plate of 1/8 inch thick section; view #6 of ASTM E192.

3.3.2.1.3 The total indications (smooth edged and/or irregularly shaped) in Section A, shall not exceed the total number of indications or aggregate area of all indications shown in the gas hole plate or 1/8 inch thick section; view #6 of ASTM E192.

3.3.2.2 Section B (see dwg. 9207928, covering a detail of dwg. 9241480).

3.3.2.2.1 No individual smooth edged indication, in Section B, shall exceed the area of the largest hole depicted in the gas hole plate for 3/8 inch thick section; view #6 of ASTM E192.

3.3.2.2.2 No individual irregularly shaped indication, in Section B, shall exceed the area of the largest indication depicted in the foreign material less dense plate of 3/8 inch thick section; view #7 of ASTM E192.

3.3.2.2.3 The total indications (smooth edged and/or irregularly shaped), in Section B, shall not exceed the total number of indications or aggregate area of all indications shown in the gas hole plate for 3/8 inch thick section; view #6 of ASTM E192.

3.3.2.3 No projected area, in Section A or B, which can be included in a circle 1/2 inch in diameter, shall have more than 5 (smooth edged and/or irregularly sahped) individual indications.

3.3.2.4 Indications which may be interpreted as smooth edged indications, as opposed to irregularly shaped indications, shall be so categorized and the applicable requirements for smooth edged indications shall apply.

3.4 Security.-There shall be no evidence of breakup of the projectile in the gunbore or in flight when examined after recovery, when tested in accordance with 4.4.5.

3.5 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship.-The projectiles shall be fabricated and finished in a thorough, workmanlike manner. They shall be free

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from dirt, grease, chips, oil, rust, corrosion or other foreign matter . Particular attention shall be given to the cavity of the body, prior to painting, to insure that no grit, scale, burrs, or other foreign matter remains therein after final cleaning. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The intial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

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4.2.1 Submission.-Prior to initiation of regular production, the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 35 projectiles (25 unpainted and 10 painted) which have been produced by the contractor or furnished by a supplier, and which have been manufactured using the same production process, procedures and equipment which will be used in fulfilling the contract. In addition, the contractor shall submit a sample of 50 projectile assemblies for ballistic testing in accordance with 4.4.5. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of the inspection with the sample, including statements of findings for materials, processes used and tests. A sample containing known defects will not be submitted, unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Parts will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification, and any or all requirements of the applicable drawings. All tests shall be performed as specified in 4.4.

4.2.3 Rejection.-If any part fails to comply with any of the applicable requirements, for the inspections performed in accordance with 4.3, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any or all of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quanitities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and

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same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235, may be used if approved by the Government. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Body (see dwg. 9207928, covering a detail of dwg. 9241480).

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:		
101.	Pitch diameter of internal thread at fuze end, maximum (max.)	Gage (CD)
102.	Pitch diameter of internal thread at fin end, max	Gage (CD)
103.	Minor diameter of internal thread at fuze end, max	Gage (CD)
104.	Minor diameter of internal thread at fin end, max	Gage (CD)
105.	Front threads must accept a minimum (min.) pitch diameter thread gage	Gage (CD)
106.	Rear thread cavity must receive a min. pitch diameter thread gage for .39 min. length	Gage (CD)
107.	Length of internal thread at fuze end, min	Gage (CD)
108.	Depth of fin thread cavity, max	Gage (CD)
109.	Depth of internal cavity	Gage (CD)
110.	Wall thickness at bourrelet, min	Gage (CD)

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111. Wall thickness at rear basic diameter, min....Gage (CD)
112. Wall thickness at forward basic diameter, min.....Gage (CD)
113. Diameter of bourrelet, min.....Gage (CD)
114. Depth of grooves in bourrelet.....Gage (CD)
115. True position of fuze thread with bourrelet...Gage (CD)
116. True position of fin thread with bourrelet....Gage (CD)
117. True position of shell taper at fin end with bourrelet.....Gage (CD)
118. True position of shell taper at fuze end with bourrelet.....Gage (CD)
119. Total length.....Gage (CD)
120. Weight.....Gage (CD)

Minor:
201. Length to shell taper at fuze end at smallest basic diameter.....Gage (CD)
202. Length to shell taper at end at smallest basic diameter.....Gage (CD)
203. Length to rear of bourrelet.....Gage (CD)
204. Diameter of body at rear of bourrelet.....Gage (CD)
205. Sharp corners missing at front and rear edges of grooves and rear edges of bourrelet.....Visual
206. Diameter of fin thread undercut, max. (if applicable).....Gage (CD)
207. Chamfer at rear missing or incorrect.....Visual-Gage (CD)
208. Chamfer at nose missing or incorrect.....Visual-Gage (CD)
209. Surface finish improper.....Visual
210. Evidence of poor workmanship (see 3.7).....Visual

4.3.2.2 Body (Marking) (see dwg. 9241480).

Categories	Defects	Method Of Inspection
Critical:		
1.	Pool of paint in cavity not solidified or containing void	Visual-Tactile
2.	Air pressure, hydrostatic or magnetic particle inspection stamp missing or illegible	Visual
Major:	100 percent inspection	
101.	Surface defects in excess of allowable criteria after painting (see 3.3)	Visual
Major:		
102.	Diameter of bourrelet max	Gage (CD)

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103.	Threads damaged or not full for entire circumference	Visual
Minor:		
201.	Exterior paint not dry	Tactile
202.	Interior paint not dry	Visual-Tactile
203.	Exterior coating with bare spot in excess of 1/4 square inch	Visual
204.	Interior coating inadequate (bare spot or scratch)	Visual
205.	Paint on nose end, base end or threads	Visual
206.	Light coat of corrosion preventive compound not present	Visual-Tactile

4.3.2.3 Carton, prior to sealing (see dwg. 7548319).

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:	None defined.	
Minor:		
201.	Support missing or damaged	Visual
202.	Separator missing or damaged	Visual
203.	Projectile missing	Visual

4.3.2.4 Carton, after sealing (see dwg. 7548319).

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:	None defined.	
Minor:		
201.	Carton damaged, cut or gouged through all layers	Visual
202.	Carton not completely sealed	Visual
203.	Contents loose	Visual-Manual
204.	Marking missing, misleading or unidentifiable	Visual

4.3.3 Testing

4.3.3.1 Mechanical properties

4.3.3.1.1 Hardness (see dwg. 9207928, covering a detail of dwg. 9241480).-In order to insure uniformity of heat-treatment and to establish a hardness range, to assure compliance with the mechanical properties of dwg. 9207928, castings of the heat-treatment lot shall be subjected to this test performed as specified in 4.4.1.1 using equipment in accordance with 4.3.4.

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4.3.3.1.1.1 Beginning with the first lot produced by each foundry and continuing until castings numbering no less than 100 consecutive final heat-treatment lots each comprised of four hours production have complied with the acceptance criteria specified, 100 percent hardness testing shall be performed. The contractor shall maintain and submit to the Government, for informational purposes of the technical agency only (see 6.7), a history of the final heat-treatment for those periods requiring 100 percent hardness testing. Lot data shall include lot number, total number of pieces in the lot, frequencies of brinell hardness values, and results of tension tests as prescribed by 4.3.3.1.2.

4.3.3.1.1.2 After 100 percent hardness testing criteria of 4.3.3.1.1.1 have been met, the hardness test shall be performed on a randomly selected sample representing 10 percent but not less than 400 castings of the final heat-treatment lot. When reduced hardness testing is in effect, 100 percent testing specified in 4.3.3.1.1.1 shall be instituted if any of the following occur:

a. Three lots are rejected for failure to meet mechanical properties within any 25 calendar day period.
b. Any major change of foundry technique, such as risering, chilling, sand mix or pouring temperatures.
c. Start of a new foundry or furnance. (Includes lapse of production of 3 months or more).

4.3.3.1.2 Tension test (see dwg. 9207928, covering a detail of dwg. 9241480), Major Defect.-Two castings from each lot representing the hardest and softest castings as determined by the hardness test outlined in 4.3.3.1.1 shall be subjected to the tension test.
Two specimens from each casting shall be prepared as specified in 4.4.1.2 and tested. If all specimens comply with the yield strength and elongation requirements of dwg. 9207928, the hardness range for the lot will have been established. Failure of one or more tension test specimens to comply with the requirements shall be cause for rejection of the lot represented by the specimen.

4.3.3.1.3 Retest.-When one or more tension test specimens fail to comply with the yield strength and elongation requirements of dwg. 9207928 and do not qualify for retest under 4.4.1.2.3, a new hardness range shall be established. The rejected lot may, at the discretion of the contractor, be reheat-treated prior to establishing a new hardness range. The tension test, in compliance with 4.3.3.1.2, shall be conducted to determine the acceptability of the new hardness range. When the hardness range has been established, all castings whose values fall outside this range shall be rejected.

4.3.3.2 Soundness

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4.3.3.2.1 Surface indications (see 3.3.1), Critical Defect.-This test shall be performed 100 percent in accordance with the procedure specified in 4.4.2.1, using equipment in accordance with 4.3.4. If any body fails to comply with the specified requirement, it shall be classified defective and removed from the lot.

4.3.3.2.2 Radiographic indications (see 3.3.2), Major Defect

4.3.3.2.2.1 Beginning with the first lot produced by each foundry and continuing until three consecutive lots have complied with the acceptance criteria specified, 500 randomly selected shell bodies which have received hydrostatic test, air test, and magnetic particle inspection shall be examined. Failure of one or more samples to meet the radiographic requirements of 3.3.2 shall be cause for rejection of the lot. If it can be determined that a radiographic indication is caused by a surface defect that would normally be visible after paint, that body should be removed from the sample and replaced with another randomly selected sample. Similarly, a second replacement body, but not more than two replacement bodies, may be randomly selected if required. The test shall be performed as specified in 4.4.2.2, using equipment in accordance with 4.3.4.

4.3.3.2.2.2 After three consecutive lots have met the criteria of 4.3.3.2.2.1, 25 randomly selected shell bodies which have received hydrostatic test, air test, and magnetic particle inspection shall be examined. Failure of one or more samples to meet the radiographic requirements of 3.3.2 shall be cause for rejection of the lot. If it can be determined that a radiographic indication is caused by a surface defect that would normally be visible after paint, that body should be removed from the sample and replaced with another randomly selected sample. Similarly, a second replacement body, but not more than two replacement bodies, may be randomly selected if required. The test shall be performed as specified in 4.4.2.2, using equipment in accordance with 4.3.4. In the event a lot is rejected, the sampling plan shall revert to the acceptance procedure specified in 4.3.3.2.2.1 until again three consecutive lots have complied with the applicable requirement.

4.3.3.3 Hydrostatic test (see dwg. 9207928, covering a detail of dwg. 9241480), Critical Defect.-This test shall be performed 100 percent in accordance with the procedure specified in 4.4.3, using equipment in accordance with 4.3.4. If any body fails to comply with the specified requirement, it shall be classified defective and removed from the lot.

4.3.3.4 Air pressure test (see dwg. 9207928, covering a detail of dwg. 9241480), Critical Defect.-Each body that has passed the hydrostatic test shall be subjected to the air pressure test as specified in 4.4.4, using equipment in

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accordance with 4.3.4. If any body fails to comply with the specified requirement, it shall be classified defective and removed from the lot.

4.3.3.5 Proving ground

4.3.3.5.1 Initial production sample.-The quantities listed in Table I shall be ballistically tested in the phases indicated. Tests shall be performed as specified in 4.4.5.

TABLE I

Phase	Quantity
Security (Critical Defect)	25
Velocity and Accuracy	25

Failure of any round fired for security to comply with the requirements of 3.4 shall be cause for rejection of the lot. Velocity and accuracy will be recorded for informational purposes only (see 6.4).

4.3.3.5.2 Production sample.-The quantities listed in Table II shall be ballistically tested in the phase indicated. Tests shall be performed as specified in 4.4.5.

TABLE II

Phase	Quantity
Security (Critical Defect)	10 each lot

Failure of any round fired for security to comply with the requirements of 3.4 shall be cause for rejection of the lot.

4.3.4 The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method Of Inspection" headings in 4.3.2 and 4.3.3 (see 6.3).

4.4 Test methods and procedures

4.4.1 Mechanical properties

4.4.1.1 Hardness.-Body castings of the heat-treatment lot shall be subjected to a Brinell hardness test using ASTM Method E10 on the bourrelet of the body casting after final treatment. Any quantity of castings not to exceed four (4) hours production of the final heat-treatment in the same

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furnance, for the same length of time or at the same conveyor speed, shall be considered as a heat-treatment lot. The lot thus constituted shall be the basis for selection of sample castings to be used for the tension tests of 4.3.3.1.2.

4.4.1.2 Tension test.-From each sample selected in accordance with 4.3.3.1.2, two tension test specimens shall be taken longitudinally from each side of the sample, 180° apart.

4.4.1.2.1 Tension test specimen.-The specimen shall be the largest obtainable standard round specimen prescribed by ASTM Method E8 (figure 8) which the size of the casting and location of the specimen will permit.

4.4.1.2.2 Yield strength and elongation.-Yield strength shall be determined by the offset method as defined in ASTM Method E8. The offset shall be 0.1 percent (0.001 inches per inch of gage length). Elongation on each specimen shall be determined by utilizing the criteria for these tests prescribed by ASTM Method E8.

4.4.1.2.3 Replacement test specimens.-If a tension test specimen fails to meet the requirements of dwg. 9207928, due to an identifiable defect in the tensile specimen which is acceptable under provisions of 3.3 and 4.3.3.2 or due to the fracture occurring under those conditions allowing for retests for elongation in ASTM Method E8, another specimen shall be machined from the same casting and tested. In the instance of failure due to an acceptable metal defect only one replacement specimen shall be permitted, and failure of the replacement specimen to comply with the requirements of 3.3 shall be cause for rejection of the represented lot. Results of all tests, original and replacement test, shall be recorded and specimens which failed to meet the requirements due to the above conditions shall be identified and retained by the contractor and made available to the procuring activity for examination upon request.

4.4.2 Soundness

4.4.2.1 Surface indications.-Magnetic particle inspection shall be performed for this test in accordance with the procedures as specified in MIL-M-11472, for indirect magnetization by either the continuous method or the residual method. The solid content of the suspension shall be maintained at 0.2 to 0.4 milliliters (ml) of solids per 100 ml of suspension when oil is used, or 0.15 to 0.35 ml of solids per 100 ml of suspension when water is used. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by section 4 of MIL-M-11472. When performing the test for solid content of the suspension, if it is determined that it does

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not comply, all body castings that have been tested and accepted since the last satisfactory test was performed shall be retested after the suspension has been brought to the proper level. The operator(s) of the equipment and the inspector(s) making the determination shall be certified in accordance with MIL-STD-410. This test shall be performed after air test and prior to painting.

4.4.2.2 Radiographic indications.-The projectile shall be radiographically inspected using equipment, operations and procedures in accordance with MIL-STD-453 and the following:

a. The sensitivity of x-ray technique shall be considered satisfactory when the image of the penetrameter, including the 2T diameter hole, is visible on the x-ray film.

b. Penetrameters shall be placed on the upper surface of each of two blocks of metal of the same alloy composition and approxaimtely the same thickness as the parts radiographed, and located on the film at an area at the outer edge of the cone of radiation. Due to varying wall thickness, the projectile is sectioned into section "A" and "B" (see dwg. 9207928). Wall thickness shall be measured at approximately 1.7 inches and 3.8 inches from the projectiles fuze end. The metal blocks shall be 3/4" x 1 1/2" x (2 times the wall thickness of the shell section radiographed). The 3/4" and 1 1/2" faces shall be normal to and within the outer edge of the cone of radiation. Thus, two penetrameters shall appear on each radiographic negative, one for each section "A" and "B" of the projectile.

c. Each projectile shall be x-rayed twice. The "H" stamping (see dwg. 9207928) shall be used as a 0° reference. X-rays shall be taken at the 0° and 90° positions.

4.4.2.2.1 Prior to radiographic inspection, the blocks, penetrameters, equipment and radiographic inspection procedure shall be approved by the responsible Government procurement agency.

4.4.3 Hydrostatic.-The body assembly shall be hydrostatically tested using suitable equipment. In performing the test, the pressure shall be applied gradually until the required minimum pressure is reached, after which the pressure shall be held for a minimum of 5 seconds. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 2 hours of continuous operations. Also, it shall be calibrated prior to restarting the operation after layoff of one hour or more.

4.4.4 Air pressure.-Body assemblies that have passed the hydrostatic test and have been dried by heating at 250° Fahrenheit (F) for a minimum of 20 minutes shall be mounted in

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an approved fixture, completely submerged in water and pressurized in accordance with applicable drawing requirement. The water in the tank shall be permitted to settle after immersion of the assemblies prior to the start of timing. Observation shall be made 360° of the periphery for the full 15 seconds the assembly is required to be timed. The equipment shall be calibrated at the start of each day's operation and at intervals of 4 hours and after a work stoppage of 3 hours or more.

4.4.5 Proving Ground.-These tests shall be conducted at a Government proving ground in accordance with the applicable Material Test Procedure and Acceptance Test Procedure.

4.4.5.1 Security.-The sample projectiles shall be inert loaded prior to firing to 3.10 ± .05 pounds, assembled into complete rounds and fired in a mortar for which the round is standard. The propellant charge shall be adjusted to obtain 6700 ± 400 pounds per square inch (psi) in a new mortar. Prior to loading the projectile, length and the outside diameter (measured at three different locations and two planes located 90° apart; bourrelet, 1 inch and 3 inches from small end) shall be measured and recorded. After recovery, the projectile shall be measured to determine compliance with 3.4. (At least 60 percent of the projectiles shall be recovered in this phase. If less than 60 percent are recovered, a retest shall be performed).

4.4.5.2 Velocity-accuracy.-The sample projectiles shall be inert loaded to the service weight, assembled into complete rounds, and fired at charge 4 in a mortar for which the round is standard. For comparison, and equal number of rounds from a reference lot shall be fired. The sample rounds and reference rounds will be fired alternately. Velocity, range, and deflection will be recorded for each round for determination of velocity, range, and deflection probable error (PE) to assess accuracy. Data obtained shall be recorded for information only and forwarded to the cognizant technical agency.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C.-The projectile shall be preserved and packaged in accordance with dwg. 7548319.

5.2 Marking.-Marking of the cartons shall be in accordance with dwg. 7548319.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

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b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs, if available, are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specifed, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, that established the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number, paragraph number and defect number if such number has been assigned shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

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6.2.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 Ballistic test.-Rounds fired for velocity are not acceptance tests, but the results will be carefully analyzed by the Armament Command to determine quality of ammunition produced. Evidence of excessive variation in either phase of the proving ground tests will result in immediate investigation by the Armament Command.

6.4 Submission of test data.-In addition to the normal distribution of records, when the cartridge is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to the following:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-A
Dover, New Jersey 07801

6.5 Definitions

6.5.1 Crack.-A discontinuity which has a relatively large cross-section in one direction and a small or negligible cross-section when viewed in a direction perpendicular to the first.

6.5.1.1 Cold cracks.-Straight or jagged lines usually continuous throughout their length. Cold cracks generally appear singly. They start at the surface. (See ASTM, E192, reference radiograph).

6.5.1.2 Hot cracks.-Appear as ragged, dark lines of variable width and numerous branches. They have no definite line of continuity and may exist in groups. They may originate internally or at the surface.

6.5.2 Cold shut (on radiograph).-A distinct dense line or band of variable length and definite smooth outline. This discontinuity appears on the surface of cast metal as a result of two streams of liquid meeting and failing to unite. (See ASTM, E192, reference radiograph).

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6.5.3 Foreign material (See ASTM, E192, reference radiograph)

6.5.3.1 Foreign material less dense (on radiograph).- Irregularly shaped indications more dense than the background but less dense than gas holes of similar magnitude.

6.5.3.2 Foreign material more dense (on radiographic).- Irregularly shaped indications less dense than the background

6.5.4 Gas holes.-Holes created by a gas escaping from molten metal appear as round or elongated, smooth edged spots, occuring individually, in clusters or distributed throughout a casting (See ASTM, E192, reference radiograph).

6.5.5 Hot tears (on radiograph).-Ragged, dark lines of variable width and numerous branches. They have no definite lines of continuity and may exist in groups. They may originate internally or at the surface. The fracture is formed in a metal during solidication because of hindered contraction. (See ASTM, E192, reference radiograph).

6.5.6 Indication.-In non-destructive inspection, a response or evidence of a response that requires interpretation to determine its significance.

6.6. Final heat-treatment hsitory.-Final heat-treatment history will be analysed by Picatinny Arsenal to evaluate production consistency and the probability with which a 10 percent sample will select castings containing both minimum and maximum hardness values existent in the lot.

Custodian:
ARMY-PA

Preparing Activity:
ARMY-PA

Project Number: 1310-A-049

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PENALTY FOR PRIVATE USE $300

Commander
Picatinney Arsenal
ATTN: SARPA-AD-E-C-3
Dover, NJ 07801

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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